

RESEARCH INTENT

WQF7023 Artificial Intelligence Research Project

Name	Yip Zi Xian
Matric No	24202141
Supervisor	Prof. Ts. Dr. Rafidah Md Noor
Area of Research	Artificial Intelligence, Deep Learning, Intelligence Transportation, Urban Informatics
Research Title	Assessing Public Transit Reliability through Deep Learning Analytics in Malaysia
Introduction	This research addresses the challenges of public transit reliability, focusing on delays and ridership trends in Malaysia. By leveraging deep learning, it aims to provide data-driven insights to support SDG 11: Sustainable Cities and Communities.
Statement of Aim	To develop and evaluate advanced deep learning models that accurately predict transit delays and ridership, identifying operational bottlenecks to improve urban transport efficiency in Malaysia.
Research Questions	1. How can deep learning models predict public transit delays and passenger ridership trends in Malaysia using GTFS? 2. What are the key factors contributing to unreliability in urban public transit networks in Malaysia? 3. How do different deep learning architectures perform compared to baseline model in predicting transit delays and ridership patterns? 4. How can insights derived from predictive models be used to improve reliability, efficiency, and service quality of urban public transportation systems in Malaysia?
Research Objectives	- Develop robust deep learning models for predicting public transit delays and forecasting passenger ridership trends in Malaysia. - Identify critical factors contributing to unreliability within urban transit networks - Evaluate the performance of various deep learning architectures against established baseline in the context of public transit reliability assessment. - Propose strategies and recommendations for improving the overall quality and efficiency of urban public transport services based on data-driven insights.
Scope	Focuses on urban bus transit networks using GTFS Static data, comparing LSTM with advanced architectures like CNN-LSTM, T-GCN, MT2CN, STGNN, and STGCN.
Significance of the Research	Addresses the knowledge gap in complex spatio-temporal transit modeling. Impacts urban planning by providing tools to enhance reliability and sustainability in alignment with SDG 11.
Methodology	A data-driven approach analyzing GTFS Static data. It involves comparative modeling using LSTM (baseline) and proposed architectures (CNN-LSTM, T-GCN, MT2CN, STGNN, STGCN) for predictive analytics.
Expected Outcomes	- Comprehensive understanding of the factors influencing public transit reliability through deep learning analytics - Validated deep learning models capable of accurately predicting transit delays and ridership - Identification of specific operational bottlenecks and peak load conditions within transit networks - Data-driven recommendations for transit authorities to enhance service efficiency, reduce delays, and improve passenger experience
Skills Needed	Python, Deep Learning framework (PyTorch), Geospatial data analysis (GTFS)
Timeline/ Gantt Chart	Phase 1: Report Documentation for Introduction, Literature Review, and Methodology Phase 2: Data Preprocessing, Model Development, Evaluation and Optimization Phase 3: Report Documentation for Discussion, Results, Future Works, and Conclusion